

A Hybrid Approach to Digital Image Watermarking: Integrating DWT, DFT, and Genetic Algorithm

By

Shahariyr Reza

ID: 11200120524



Department of Computer Science and Engineering
Northern University of Business and Technology Khulna
Khulna 9100, Bangladesh

Abstract

Image watermarking is a crucial technique for ensuring copyright protection and authentication in digital multimedia. This thesis proposes a hybrid approach to achieve secure and robust image watermarking through a combined framework utilizing Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA). The objective is to enhance the imperceptibility of the embedded watermark while maintaining robustness against various attacks such as noise addition, compression, and geometric transformations. The proposed method leverages the multi resolution capabilities of DWT for embedding the watermark in frequency sub bands and employs DFT for frequency domain analysis to strengthen watermark imperceptibility. Genetic Algorithm is integrated to optimize the embedding and extraction process, thereby improving robustness against attacks. Experimental evaluations demonstrate the effectiveness of the proposed framework in achieving high fidelity and resilience to malicious attacks compared to existing watermarking techniques. This research contributes to the advancement of secure digital watermarking techniques by introducing a comprehensive framework that combines the strengths of DWT, DFT, and GA. The results we found is an average PSNR value of 41.67 which underscore its potential applicability in various domains requiring copyright protection and content authentication in digital multimedia.

Keywords: Image watermarking, Discrete Wavelet Transform, Discrete Fourier Transform, Genetic Algorithm, security, robustness.

Contents

	PAGE
Title Page	
Declaration	i
Approval	ii
Acknowledgement	iii
Abstract	iv
Contents	v-vii
List of Tables	viii
List of Figures	ix
List of Algorithms	x
List of Equations	xi
List of Abbreviations	xii
CHAPTER I Introduction	1
1.1 Overview	1
1.2 Motivation	2
1.3 Research Question	3
1.4 Objectives	3
1.5 Thesis Organization	4
CHAPTER II Literature Review	6
2.1 Watermarking Techniques	6
2.1.1 Discrete Cosine Transform (DCT)	6
2.1.2 Discrete Wavelet Transform (DWT)	8
2.1.3 Discrete Fourier Transform (DFT)	9
2.2 Watermark Optimization Techniques	9
2.2.1 Singular Value Decomposition (SVD)	10
2.2.2 Genetic Algorithm (GA)	10

	2.2.3 Particle Swarm optimization (PSO)	12
	2.3 Used Python Library	12
	2.3.1 NumPy	13
	2.3.2 Open CV	13
	2.3.3 PyWavelets	13
	2.3.4 SciPy	13
	2.3.5 Matplotlib	13
	2.4 Performance matrix	14
	2.4.1 MSE (Mean Square Error)	14
	2.4.2 PSNR (Peak Signal to Noise Ratio)	14
CHAPTER III	Related Studies	16-24
CHAPTER IV	Methodology	25
	4.1 System Architecture	25
	4.2 Dataset	26
	4.2.1 Original Images	26
	4.2.2 Watermark Images	26
	4.3 Watermark Embedding Procedure	27
	4.3.1 Preprocessing	27
	4.3.2 Discrete Wavelet Transform (DWT)	27
	4.3.3 Discrete Fourier Transform (DFT)	28
	4.3.4 Watermark Embedding	29
	4.3.5 Genetic Algorithm (GA) Optimization	30
	4.4 Watermark Extraction	31
CHAPTER V	Results and Discussion	32
	5.1 PSNR for DWT and DFT	32
	5.2 PSNR for DWT, DFT with GA	32
	5.3 Comparison with Related Work	33
	5.4 Discussion	34

CHAPTER VI	Conclusions and Future Work	35
	6.1 Conclusion	35
	6.2 Future work	35
References		37-39

List of Tables

Table No.	Description	Page
3.1	Comprehensive analysis of previous studies.	22
5.1	The PSNR value after applying DWT and DFT.	32
5.2	The PSNR value after applying Genetic Algorithm	33
5.3	The Imperceptibility Comparison with Related Work.	33

List of Figures

Figure No.	Description	Page
2.1	Discrete Wavelet Transform (DWT)	8
2.2	Genetic Algorithm	11
4.1	System architecture for watermarking.	25
4.2	Original Images	26
4.3	Watermark Images	27
4.4	Watermark Embedding	29
4.5	Watermark Embedded Images	30
4.6	Optimized Watermark Embedded Images	31

List of Algorithms

Algorithm No.	Description	Page
4.1	Discrete Wavelet Transform (DWT)	28
4.2	Discrete Fourier Transform (DFT)	28

List of Equations

Equation No.	Description	Page
2.1	Discrete Cosine Transform (DCT)	7
2.2	Scaling Factor of (DCT)	7
2.3	Inverse Discrete Cosine Transform (IDCT)	7
2.4	Discrete Fourier Transform (DFT)	9
2.5	Complex exponential function of DFT	9
2.6	Singular Value Decomposition (SVD)	10
2.7	Particle Swarm Optimization (PSO)	12
2.8	Mean Squared Error (MSE)	14
2.9	Peak Signal to Noise Ratio	15

List of Abbreviations

DCT	Discrete Cosine Transform
IDCT	Inverse Discrete Cosine Transform
DWT	Discrete Wavelet Transform
DFT	Discrete Fourier Transform
IDFT	Inverse Discrete Fourier Transform
GA	Genetic Algorithm
MSE	Mean Square Error
PSNR	Peak Signal to Noise Ratio
SVD	Singular Value Decomposition

CHAPTER I

Introduction

1.1 Overview

Digital watermarking is a method for protecting intellectual property and copyrighted material. It is a marker embedded in digital content material, typically used to identify the source and ownership of copyrighted material. [1] Today's digital era, the rapid proliferation of digital multimedia content, particularly images, has highlighted the critical need for effective content protection and authentication mechanisms. Image watermarking has emerged as a pivotal technology in addressing these challenges by embedding imperceptible information, or watermarks, into digital images. These watermarks serve diverse purposes, ranging from copyright protection and ownership verification to tamper detection and content authentication.

The concept of digital watermarking draws its roots from traditional watermarking techniques used in paper documents, where a faint mark or pattern was added during manufacturing to assert ownership or authenticity. In the digital realm, however, the complexity and requirements of watermarking extend far beyond mere visibility or physical presence. Digital watermarks must be robust against common image processing operations, yet imperceptible to human observers to preserve the visual quality and integrity of the host image.

This thesis explores the fundamental principles, methodologies, and advancements in image watermarking techniques. It delves into both theoretical foundations and practical implementations, addressing key challenges such as robustness, security, capacity, and fidelity. By investigating various approaches including spatial domain, frequency domain, and transform domain methods, this study aims to provide a comprehensive understanding of the underlying mechanisms and their application in real-world scenarios. Moreover, the evolution of digital image watermarking intersects with broader technological advancements, including machine learning algorithms, deep neural networks, and advancements in image processing techniques. These developments have not only enriched the capabilities of watermarking systems but also introduced new challenges and opportunities for innovation.

Through critical analysis and empirical evaluations, this thesis aims to contribute to the existing body of knowledge in image watermarking, proposing novel methodologies or enhancements to existing techniques that address current limitations and anticipate future requirements in digital content protection and authentication.

In conclusion, understanding the intricacies of image watermarking is crucial for safeguarding digital content integrity and promoting trust in digital communication and commerce. This thesis endeavors to provide insights that contribute to the advancement and adoption of secure and efficient image watermarking solutions in practical applications.

1.2 Motivation

In digital world, the act of image sharing and distribution has significantly amplified the challenges associated with protecting digital content. The ease with which digital images can be copied, manipulated, and redistributed poses substantial risks to intellectual property rights, content authenticity, and data integrity. These challenges necessitate advanced solutions that can safeguard digital images against unauthorized use, tampering, and piracy.

- **Protecting Intellectual Property Rights**
The protection of intellectual property (IP) has become a pressing issue in the digital age. Artists, photographers, designers, and content creators often find their work being used without permission or proper attribution. Digital image watermarking provides a viable solution by embedding unique identifiers or signatures into images, which can serve as proof of ownership and deter unauthorized usage [2]. By developing a robust watermarking method, this research aims to contribute to the enforcement of IP rights, ensuring that creators receive recognition and compensation for their work.
- **Ensuring Content Authenticity and Integrity**
In various fields such as journalism, scientific research, and legal investigations, the authenticity and integrity of digital images are paramount. Tampered images can lead to misinformation, fraudulent activities, and compromised evidence. A reliable watermarking method can act as a tamper-proof seal, enabling users to verify whether an image has been altered. This capability is crucial for maintaining the trustworthiness of digital visual content in critical applications.
- **Addressing Limitations of Existing Methods**
While numerous watermarking techniques have been developed, we may face many problems for robustness, imperceptibility, and computational efficiency. Traditional methods often fall to providing comprehensive solutions that are resilient to various image processing attacks while maintaining high visual quality. By integrating Discrete Wavelet Transform (DWT) and Discrete Fourier Transform (DFT), and optimizing the process using Genetic Algorithm (GA), this research seeks to overcome these limitations, providing a more effective and balanced approach to digital image watermarking.
- **Contributing to the Advancement of Digital Content Protection**
The advancement of digital content protection technologies is essential for fostering a secure and trustworthy digital environment. By developing a novel watermarking method, this research aims to contribute to the broader efforts in enhancing digital security and privacy. The outcomes of this thesis have the potential to impact

various sectors, including media, entertainment, healthcare, and forensics, by providing a reliable tool for protecting digital images.

The motivation for this thesis is driven by the urgent need to protect digital images, the opportunity to address current limitations in watermarking methods, the potential of advanced computational techniques, and the desire to contribute to the overall enhancement of digital content security. This research aspires to develop a robust, imperceptible, and secure watermarking method that meets the evolving demands of digital content protection.

1.3 Research Question

How can a hybrid digital image watermarking method, integrating Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA), be designed and optimized to achieve high robustness, imperceptibility, and security against common image processing attacks and manipulations?

To address this primary research question, the following sub-questions are formulated:

- How can the Discrete Wavelet Transform (DWT) and Discrete Fourier Transform (DFT) be effectively combined to enhance the robustness and imperceptibility of digital image watermarks?
- In what ways can a Genetic Algorithm (GA) be employed to optimize the watermark embedding and extraction processes?
- How can the imperceptibility of the watermark be maintained without compromising the visual quality of the host image?
- How does the proposed hybrid method compare with existing digital image watermarking techniques in terms of robustness, imperceptibility, capacity, and computational efficiency?
- How can a practical and user-friendly tool be developed to implement the proposed watermarking method for real-world applications?

1.4 Objectives

The primary objective of this thesis is to develop a robust and efficient digital image watermarking method by integrating Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA). This integrated approach aims to enhance the robustness, imperceptibility, and security of the watermarking process. The specific aims of this research are outlined as follows:

Develop a comprehensive watermarking framework that leverages the strengths of DWT and DFT for embedding watermarks into digital images. Implement a Genetic Algorithm to optimize the watermarking process, ensuring an optimal balance between robustness and imperceptibility. Evaluate the resistance of the proposed watermarking method against common image processing attacks such as compression, cropping, noise addition, and geometric transformations. Fine-tune the parameters of DWT, DFT, and GA to maximize the robustness of the embedded watermark under various attack scenarios. Ensure that the

embedded watermark remains imperceptible to human visual perception, maintaining the visual quality of the host image. Conduct subjective and objective assessments of the watermarked images to verify that the watermarking process does not degrade image quality. Develop efficient algorithms for embedding and extracting watermarks using DWT, DFT, and GA, aiming for low computational complexity and fast processing times. Validate the effectiveness of the embedding and extraction processes through extensive testing and performance evaluation. Increase the watermark capacity without compromising the robustness and imperceptibility, allowing for the embedding of larger amounts of information. Implement security measures to protect the watermark from unauthorized detection and removal, enhancing the overall security of the watermarking system. Perform comparative analysis with existing watermarking methods to demonstrate the advantages and improvements achieved by the proposed hybrid approach. Benchmark the proposed method against state-of-the-art techniques in terms of robustness, imperceptibility, capacity, and computational efficiency. Create a user-friendly software tool that implements the proposed watermarking method, facilitating its use in real-world applications. Provide documentation and guidelines for users to effectively utilize the watermarking tool for various digital content protection scenarios. By achieving these objectives, this research aims to contribute a novel and practical solution to the field of digital image watermarking, addressing current challenges and providing a robust, imperceptible, and secure method for protecting digital images in various applications.

1.5 Thesis Organization

This thesis is organized into several chapters, each focusing on a different aspect of the development and evaluation of a digital image watermarking method using Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA). The structure of the thesis is as follows:

- Chapter 1: Introduction

This chapter provides an overview of the thesis, including the background and significance of digital image watermarking. It outlines the motivation for the research, defines the research objectives and questions, and presents the scope of the study.

- Chapter 2: Literature Review

This chapter reviews existing literature on digital image watermarking techniques. It discusses the principles and applications of DWT, DFT, and GA in watermarking, highlighting their individual strengths and limitations. This review establishes the foundation for the proposed hybrid method.

- Chapter 3: Related Studies

In this chapter, related studies are examined to contextualize the research within the broader field of digital image watermarking. It provides a comparative analysis of various approaches and identifies gaps in current methodologies that the proposed method aims to address.

- Chapter 4: Methodology

This chapter details the methodology of the proposed watermarking method. It describes the integration of DWT, DFT, and GA, explaining the algorithm for watermark embedding and extraction. The chapter also discusses the optimization process using GA to balance robustness and imperceptibility, along with the experimental setup and evaluation metrics.

- Chapter 5: Results and Discussion

This chapter presents the experimental results of the proposed method. It includes quantitative and qualitative analyses of the method's performance in terms of robustness, imperceptibility, and capacity. The results are compared with existing techniques to demonstrate the improvements achieved. Additionally, the chapter discusses the implications of the findings.

- Chapter 6: Conclusion and Future Work

The final chapter summarizes the key contributions and findings of the thesis. It discusses the limitations of the current study and suggests directions for future research to further enhance the watermarking method and explore new applications.

CHAPTER II

Litarature Review

2.1 Watermarking Techniques

Image watermarking is a fundamental technique in digital media for embedding imperceptible data into images to ensure authenticity, integrity, and ownership verification. Over the years, various methods have been developed to address the challenges of robustness against attacks such as compression, cropping, and noise addition, while maintaining high perceptual quality [3].

Early techniques primarily focused on spatial domain methods like Least Significant Bit (LSB) insertion and spread spectrum techniques. LSB embedding directly modifies the least significant bits of pixel values, while spread spectrum techniques spread the watermark signal across the entire image spectrum, enhancing robustness but potentially compromising image quality. In signal processing and transform domains, frequency domain techniques such as Discrete Fourier Transform (DFT), Discrete Cosine Transform (DCT), and Discrete Wavelet Transform (DWT) gained popularity. These transforms allow for embedding watermarks in transformed coefficients, offering advantages in terms of robustness and capacity. DFT-based methods exploit the frequency spectrum of images, while DCT is commonly used in JPEG compression and watermarking applications. DWT provides multi-resolution analysis, allowing watermarks to be embedded in different frequency bands, thereby enhancing resistance to various attacks.

Furthermore, optimization techniques like Genetic Algorithms (GA) and Singular Value Decomposition (SVD) have been integrated into watermarking frameworks to enhance robustness and imperceptibility. GA optimizes watermark embedding parameters dynamically, while SVD-based methods modify singular values or vectors of images to embed watermarks effectively.

Recent research has focused on hybrid approaches that combine multiple transforms and optimization techniques to achieve superior performance in terms of robustness, imperceptibility, and capacity. These hybrid methods leverage the strengths of each component to mitigate their individual limitations and cater to diverse application scenarios.

2.1.1 Discrete Cosine Transform (DCT)

Discrete Cosine Transform (DCT) is widely used in image compression (e.g., JPEG) and has been adapted for watermarking applications. DCT-based methods embed watermarks in the frequency coefficients of images, exploiting the energy compaction property of DCT. Research by Cox et al. (1997) introduced Spread Spectrum Watermarking in the DCT domain, pioneering robust watermarking techniques in compressed domains [4].

The Discrete Cosine Transform (DCT) is an extension of the DCT applied to two-dimensional signals, such as images. It transforms the image data from the spatial domain (pixel intensity values) to the frequency domain. The DCT is widely used in image processing, particularly in image compression algorithms like JPEG, due to its ability to concentrate most of the image's energy into a few low-frequency components. Here is the mathematical definition of DCT. For an $N \times N$ image block $f(x, y)$ where x and y represent the pixel coordinates, the DCT is defined as:

$$F(u, v) = \frac{1}{4} \alpha(u) \alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos \left[\frac{(2x+1)u\pi}{2N} \right] \cos \left[\frac{(2y+1)v\pi}{2N} \right]$$

Eq 2.1: Discrete Cosine Transform (DCT)

Where:

- $F(u, v)$ is the DCT coefficient at position (u, v) ,
- $F(x, y)$ is the pixel value at position (x, y)
- N is the size of the image block (commonly 8 for 8x8 blocks in JPEG),
- u and v are the frequency coordinates,
- $\alpha(u)$ and $\alpha(v)$ are scaling factors defined as:

$$\alpha(u) = \begin{cases} \frac{1}{\sqrt{N}} & \text{if } u = 0 \\ \sqrt{\frac{2}{N}} & \text{if } u > 0 \end{cases}$$

$$\alpha(v) = \begin{cases} \frac{1}{\sqrt{N}} & \text{if } v = 0 \\ \sqrt{\frac{2}{N}} & \text{if } v > 0 \end{cases}$$

Eq 2.2: Scaling Factor of (DCT)

To reconstruct the original image block from its DCT coefficients, the inverse DCT (IDCT) is used:

$$f(x, y) = \frac{1}{4} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} \alpha(u) \alpha(v) F(u, v) \cos \left[\frac{(2x+1)u\pi}{2N} \right] \cos \left[\frac{(2y+1)v\pi}{2N} \right]$$

Eq 2.3: Inverse Discrete Cosine Transform (IDCT)

2.1.2 Discrete Wavelet Transform (DWT)

The Discrete Wavelet Transform (DWT) is a widely used technique in signal processing and image analysis. It allows for the decomposition of a signal into its constituent frequency components, enabling the analysis of both time and frequency information simultaneously. This makes it particularly useful for applications where localized frequency information is important, such as image processing, data compression, and feature extraction. [5]

The Discrete Wavelet Transform (DWT) is a mathematical technique used to break down a signal or image into different frequency components while preserving spatial or temporal information. This makes it highly useful in applications where both frequency and location data are important, such as image processing, data compression, and signal analysis. The Discrete Wavelet Transform (DWT) figure showing below in figure 2.1.

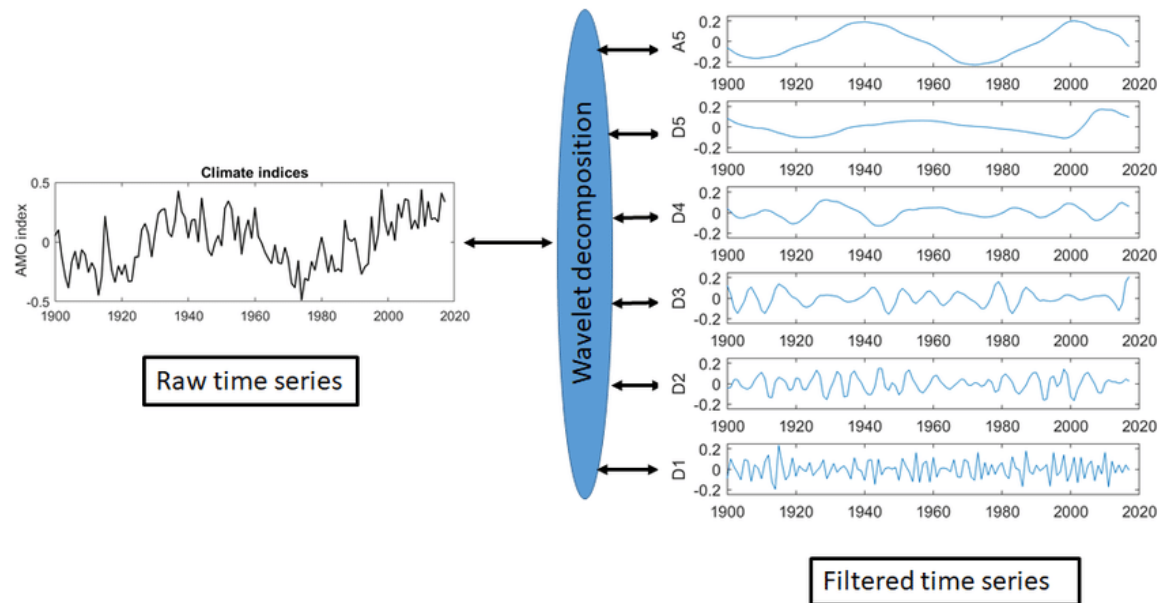


Figure 2.1: Discrete Wavelet Transform (DWT)

For applying Discrete Wavelet Transform (DWT) there is basically 4 sub bands. LL, LH, HL, HH. If any image has “A” number of row and “B” number of columns, then sub bands can be expressed as:

- $LL = A_{row} \times A_{column}$
- $LH = A_{row} \times B_{column}$
- $HL = B_{row} \times A_{column}$
- $HH = B_{row} \times B_{column}$

Where A_{row} and B_{row} are the approximation and detail coefficients obtained from the rows, and A_{column} and B_{column} are from the columns.

2.1.3 Discrete Fourier Transform (DFT)

Discrete Fourier Transform (DFT) transforms images into their frequency domain representation, offering advantages in watermark embedding through spectral manipulation [6]. The Discrete Fourier Transform (DFT) is a mathematical technique used to transform a sequence of discrete values (a signal) into its constituent frequencies. Unlike the continuous Fourier Transform, which operates on continuous signals, the DFT is designed for discrete signals, making it especially useful in digital signal processing (DSP) and image processing. The DFT converts a signal from the time (or spatial) domain into the frequency domain. This allows us to analyze the frequency components present in the signal. A discrete signal $x[n]$ of length N , the DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j \frac{2\pi}{N} kn}$$

Eq 2.4: Discrete Fourier Transform (DFT)

Here $X[k]$ is the DFT coefficient for the k -th frequency component. $x[n]$ is the original signal at the n -th sample, N is the total number of samples in the signal, k ranges from 0 to $N-1$ and represent the frequency index, j is the imaginary unit ($j^2 = -1$), $e^{-j2\pi kn/N}$ represents the complex exponential function, which can also be expressed using Euler's formula:

$$e^{-j \frac{2\pi}{N} kn} = \cos\left(\frac{2\pi}{N} kn\right) - j \sin\left(\frac{2\pi}{N} kn\right)$$

Eq 2.5: Complex exponential function of DFT

The DFT is a fundamental tool in digital signal processing, enabling the analysis and manipulation of signals in the frequency domain. By decomposing a signal into its frequency components, the DFT provides insight into the signal's behavior and allows for various processing techniques such as filtering, compression, and watermarking.

2.2 Watermark Optimization Techniques

Digital image watermarking involves embedding hidden information within an image to protect intellectual property, verify authenticity, and ensure content integrity. Optimization algorithms play a critical role in enhancing the performance of watermarking techniques by improving robustness, imperceptibility, and capacity. Some of the key optimization algorithms used in digital image watermarking include Genetic Algorithm (GA) and Singular Value Decomposition (SVD). Optimization algorithms such as GA and SVD

significantly enhance digital image watermarking by addressing the trade-offs between robustness and imperceptibility. GA provides a flexible and adaptive approach to optimize watermarking parameters dynamically, while SVD offers stability and resistance to various image processing attacks. Combining these algorithms with other techniques results in robust and efficient watermarking solutions that meet the demands of modern digital content protection.

2.2.1 Singular Value Decomposition (SVD)

Singular Value Decomposition (SVD) is a linear algebra technique that decomposes a matrix into three other matrices, revealing its intrinsic properties [7]. In digital image watermarking, SVD is used to embed watermarks by modifying the singular values of an image's matrix representation.

Singular Value Decomposition (SVD) is a fundamental technique in linear algebra used for matrix factorization. It plays a critical role in various applications such as image compression, signal processing, and machine learning. SVD decomposes a given matrix A into three distinct matrices, allowing for an insightful analysis of its structure. Given a A of size $m \times n$, the SVD can be expressed as:

$$A=U\Sigma V^T$$

Eq 2.6: Singular Value Decomposition (SVD)

U is an $m \times m$ orthogonal matrix, whose columns are called the left singular vectors of A , Σ is an $m \times n$ diagonal matrix, with non-negative real numbers on the diagonal. These numbers are known as the singular values of A , and they are ordered in descending magnitude. V^T (the transpose of V) is an $n \times n$ orthogonal matrix, whose columns are called the right singular vectors of A .

SVD-based watermarking methods often involve combining SVD with other transforms like Discrete Wavelet Transform (DWT) or Discrete Cosine Transform (DCT) to further enhance the robustness and imperceptibility of the watermark.

2.2.2 Genetic Algorithm (GA)

Genetic Algorithm (GA) is an evolutionary optimization technique inspired by natural selection and genetics. It is particularly effective in solving complex optimization problems with large search spaces, making it well-suited for digital image watermarking. The primary goal of using GA in watermarking is to find an optimal balance between robustness (resistance to attacks) and imperceptibility (minimal impact on image quality).

A Genetic Algorithm (GA) is an optimization method inspired by natural selection [7]. It starts with an initial population of candidate solutions, represented as chromosomes (often encoded as binary strings or other forms). Each chromosome is evaluated using a fitness function, which quantifies its performance on the problem. Based on fitness, better solutions are selected to create the next generation through genetic operators like crossover

(combining genetic material of two parents) and mutation (randomly altering genes to introduce variation). Crossover is often done using single-point or two-point strategies, while mutation flips bits with a certain probability. The process repeats over generations, searching for an optimal or satisfactory solution. The main operations can be summarized with equations:

1. **Fitness Calculation:** Evaluate the fitness $f(x)$ of each chromosome x .
2. **Selection:** Choose chromosomes based on fitness.
3. **Crossover:** Produce offspring from two parents using a crossover point.
4. **Mutation:** Randomly alter genes with a mutation probability that ensure diversity.

This iterative process allows GA to explore a wide solution space and converge on an optimal solution. Here is the general working flowchart of GA algorithm below.

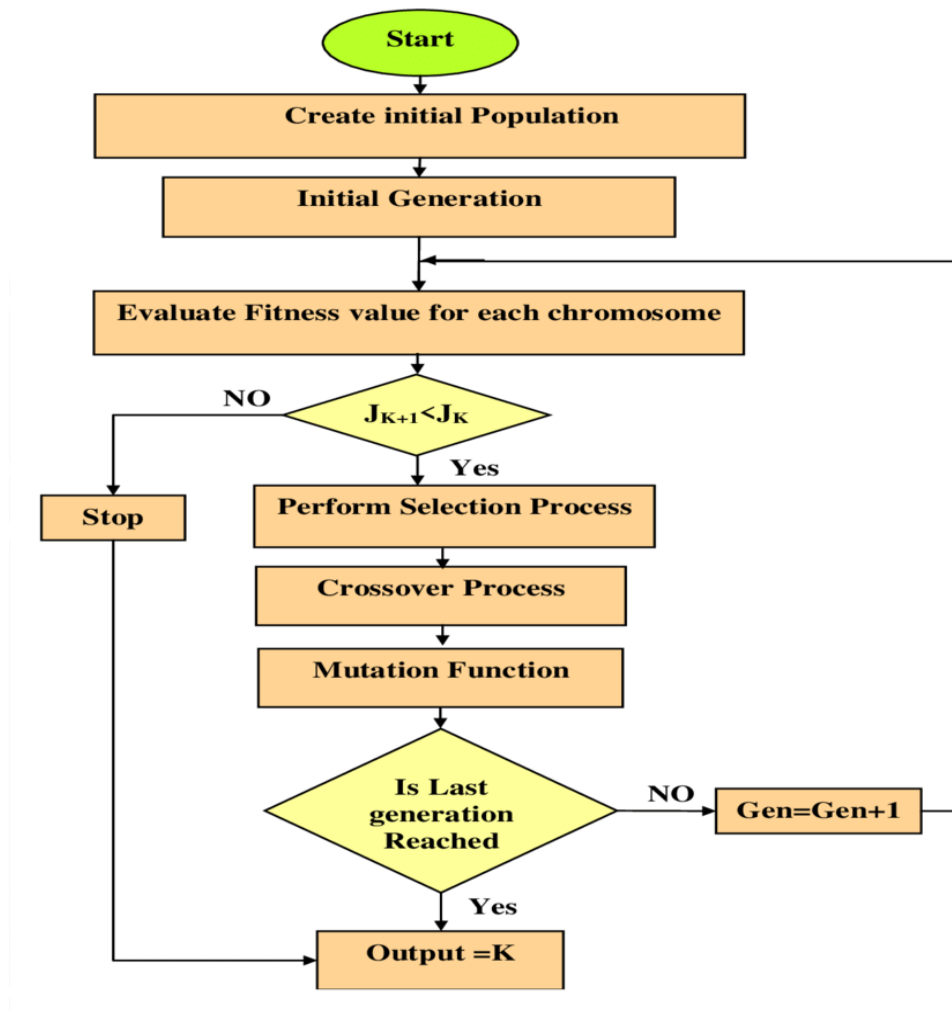


Figure 2.2: Genetic Algorithm

2.2.3 Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is an optimization technique inspired by social behaviors of animals, such as birds flocking or fish schooling, where a group of candidate solutions, called particles, move through the search space to find the optimal solution. Each particle adjusts its position based on its own best-known position (personal best, $pBest$) and the best-known position of the swarm (global best, $gBest$). The velocity update for each particle is given by [8]:

$$v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (pBest_i - x_i(t)) + c_2 \cdot r_2 \cdot (gBest - x_i(t))$$

Eq 2.7: Particle Swarm Optimization (PSO)

where $v_i(t)$ is the velocity of particle i at time t , $x_i(t)$ is its position, w is the inertia weight, c_1 and c_2 are cognitive and social coefficients, and r_1 and r_2 are random numbers between 0 and 1. The position update is then:

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

In image watermarking, PSO is used to optimize the embedding process. It helps determine the best locations and strength for embedding the watermark into the host image, ensuring minimal distortion while maximizing resistance to attacks like noise, compression, or tampering. PSO's global search capability allows it to efficiently navigate through large solution spaces, finding optimal watermark parameters in a reasonable amount of time. By leveraging PSO, watermarking systems can achieve an ideal balance between imperceptibility and robustness, making it a popular technique in modern digital watermarking methods.

2.3 Used Python Library

In this methodology, several Python libraries were utilized to accomplish various computational and visualization tasks to implement digital image watermarking [9]. NumPy was used for efficient numerical computations and manipulation of multi-dimensional arrays, essential for handling image data. OpenCV provided extensive functions for image processing, including reading, transforming, and manipulating images for watermark embedding and extraction. PyWavelets enabled the application of discrete wavelet transforms, facilitating image decomposition for watermarking. SciPy offered advanced mathematical functions and optimization algorithms crucial for refining the watermarking process. Matplotlib was employed to visualize and present the results, helping to analyze the effectiveness of the watermarking technique through plots and figures. Here is some wide details about the used Python library.

2.3.1 NumPy

NumPy is a powerful Python library used for numerical computing and data manipulation. It provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays efficiently. With its ability to perform element-wise operations, linear algebra, and statistical calculations, NumPy is an essential tool for scientific computing. Its intuitive syntax and integration with other scientific libraries like SciPy and Matplotlib make it a popular choice among data scientists and engineers for tasks ranging from simple arithmetic operations to complex simulations.

2.3.2 Open CV

OpenCV (Open Source Computer Vision Library) is a popular Python library used for computer vision and image processing tasks. It provides a comprehensive set of tools and functions for real-time image and video processing, including object detection, facial recognition, image filtering, and edge detection. OpenCV supports a wide range of file formats and can interface with various input devices like cameras. Its integration with other libraries such as NumPy allows for efficient handling of image data as arrays, making it an essential library for building computer vision applications in fields like robotics, surveillance, and augmented reality.

2.3.3 PyWavelets

PyWavelets is a versatile Python library for performing wavelet transformations, a powerful technique used in signal processing and image analysis. It provides a wide range of functions to compute one-dimensional and multi-dimensional discrete wavelet transforms (DWT), as well as continuous wavelet transforms (CWT). PyWavelets supports a variety of wavelet families, making it suitable for applications like image compression, denoising, and feature extraction. Its seamless integration with NumPy arrays enables efficient processing of large datasets, making PyWavelets an essential tool for researchers and developers working in the fields of data analysis and signal processing.

2.3.4 SciPy

SciPy is a robust Python library that builds on NumPy, providing advanced scientific and technical computing capabilities. It includes modules for optimization, integration, interpolation, eigenvalue problems, signal and image processing, and other complex mathematical tasks. SciPy's extensive functionality makes it a go-to library for researchers and engineers working on numerical algorithms and scientific applications. With its user-friendly interface and efficient algorithms, SciPy enables users to perform high-level computations and data analysis with ease. Its integration with other scientific libraries like NumPy and Matplotlib makes it an essential part of the scientific Python ecosystem.

2.3.5 Matplotlib

Matplotlib is a widely-used Python library for creating static, interactive, and animated visualizations in a variety of formats. It offers an extensive range of plotting functions,

allowing users to generate line plots, bar charts, scatter plots, histograms, and more with just a few lines of code. Matplotlib's flexibility and customization options make it a go-to tool for data visualization in scientific research, engineering, and data analysis. It integrates seamlessly with NumPy and other scientific libraries, enabling users to visualize data directly from arrays or data frames. Whether for quick data exploration or publication-quality figures, Matplotlib is an essential library for any Python data scientist or analyst.

2.4 Performance Metrics

Quality is a very important parameter for all objects and their functionalities. In image-based object recognition, image quality is a prime criterion. For authentic image quality evaluation, ground truth is required. But in practice, it is very difficult to find the ground truth. Usually, image quality is being assessed by full reference metrics, like MSE (Mean Square Error) and PSNR (Peak Signal to Noise Ratio).

2.4.1 MSE (Mean Square Error)

MSE is the most common estimator of image quality measurement metric. It is a full reference metric and the values closer to zero are the better. It is the second moment of the error. The variance of the estimator and its bias are both incorporated with mean squared error. The MSE is the variance of the estimator in case of unbiased estimator. It has the same units of measurement as the square of the quantity being calculated like as variance. The MSE introduces the Root-Mean-Square Error (RMSE) or Root-Mean-Square Deviation (RMSD) and often referred to as standard deviation of the variance.

The MSE can also be said the Mean Squared Deviation (MSD) of an estimator. Estimator is referred as the procedure for measuring an unobserved quantity of image. The MSE or MSD measures the average of the square of the errors. The error is the difference between the estimator and estimated outcome. It is a function of risk, considering the expected value of the squared error loss or quadratic loss [10].

Mean Squared Error (MSE) between two images such as $g(x,y)$ and $\hat{g}(x,y)$ is defined as

$$\text{MSE} = \frac{1}{MN} \sum_{n=0}^M \sum_{m=1}^N [\hat{g}(n, m) - g(n, m)]^2$$

Eq 2.8: Mean Squared Error (MSE)

2.4.2 PSNR (Peak Signal to Noise Ratio)

PSNR is used to calculate the ratio between the maximum possible signal power and the power of the distorting noise which affects the quality of its representation [11]. This ratio between two images is computed in decibel form. The PSNR is usually calculated as the logarithm term of decibel scale because of the signals having a very wide dynamic range.

This dynamic range varies between the largest and the smallest possible values which are changeable by their quality.

The Peak signal-to-noise ratio is the most commonly used quality assessment technique to measure the quality of reconstruction of lossy image compression codecs. The signal is considered as the original data and the noise is the error yielded by the compression or distortion. The PSNR is the approximate estimation to human perception of reconstruction quality compared to the compression codecs.

PSNR is expressed as:

$$PSNR=10\log_{10}(peakval^2/MSE)$$

Eq 2.9: Peak Signal to Noise Ratio

Here, peakval (Peak Value) is the maximal in the image data. If it is an 8-bit unsigned integer data type, the peakval is 255. we can see that it is a representation of absolute error in dB.

CHAPTER III

Related Studies

Image watermarking is a crucial technique for protecting digital media from unauthorized use and ensuring intellectual property rights. Numerous studies have explored the use of Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithms (GA) for this purpose. DWT is favored for its ability to provide both spatial and frequency domain information, making it robust against various types of image processing attacks. DFT, on the other hand, is known for its robustness against geometric distortions due to its global frequency domain representation. Genetic Algorithms are employed to optimize the embedding and extraction processes, enhancing the robustness and imperceptibility of the watermark. Studies combining these methods have demonstrated improved performance in terms of robustness, imperceptibility, and computational efficiency, making them promising for advanced image watermarking applications.

In a study, B. Kulkarni et al. proposes an Optimal SVD-DWT based Digital watermarking technique aimed at safeguarding intellectual property by preventing unauthorized copying or reproduction. The research includes a comparative survey of optimization algorithms—GA (Genetic Algorithm), PSO (Particle Swarm Optimization), and ABC (Artificial Bee Colony)—to enhance the robustness and imperceptibility of the watermarking process [12]. Evaluation metrics such as Peak Signal to Noise Ratio (PSNR), Structural Similarity Measurement (SSIM), and Normalized Cross Correlation (NCC) are employed to assess performance. The experiments utilize Python Anaconda, and computational efficiency among the optimization algorithms is quantitatively analyzed. Additionally, to mitigate false positives in SVD-based watermarking, the study integrates DWT-SVD based hashing for authentication purposes.

C.Mingzh et al. said in their study Channel estimation is crucial in OFDM systems, but conventional DFT-based methods struggle with noise reduction due to energy spreading from virtual subcarriers [13]. To improve this, a segmental threshold-assisted DFT-based algorithm is introduced, setting different thresholds based on channel and noise characteristics under varying SNR conditions, which enhances performance but still falls short at high SNR. To address this, a joint DFT and DWT channel estimation algorithm is proposed, dynamically selecting the optimal method without prior information.

M.Cedillo-Hernandez et al. proposes an optimized robust watermarking algorithm using discrete Fourier transform and spread spectrum, with key parameters automatically tuned via particle swarm optimization [14]. Experimental results show enhanced robustness against signal processing and geometric distortions while maintaining high visual quality. The proposed method outperforms conventional discrete Fourier transform and state-of-the-art particle swarm optimization techniques in image watermarking.

T. Takore et al. [15] describe that Hybrid domain digital image watermarking scheme can be used by discrete Wavelet Transform (DWT) and Singular Value decomposition (SVD). Original host image is decomposed using DWT to obtain coefficients of approximation sub-band (LL3) and it is selected for watermark insertion. Singular values of watermark image are obtained using SVD operation and scaling factor is used to modify the coefficients before inserting it into LL3 sub-band. Genetic Algorithm (GA) optimizes imperceptibility and robustness by finding best scaling factor and searching apposite location for watermark insertion. Performance measuring metric, PSNR and NCC are used to evaluate imperceptibility and robustness of the watermarked image. Experimental results obtained from simulation prove that the proposed new method is more secure and robust compared to existing approaches.

In a paper, P. Surekha et al. proposes a new optimization method for digital images in the Discrete Wavelet Transform (DWT) domain. Digital image watermarking has proved its efficiency in protecting illegal authentication of data. The amplification factor of the watermark is the significant parameter that helps in improving the perceptual transparency and robustness against attacks [16]. The tradeoff between the transparency and robustness is considered as an optimization problem and is solved by applying Genetic Algorithm. The experimental results of this approach prove to be secure and robust to filtering attacks, additive noise, rotation, scaling, cropping and JPEG compression. The Peak Signal to Noise Ratio (PSNR), Mean Square Error (MSE), and computational time are evaluated for a set of images obtained from the Tampere University of Technology, Finland using the MATLAB R2008b software.

Again, A. Boucetta et al. describes a color image compression technique based on Discrete Wavelet Transform (DWT) and Genetic Algorithm (GA). High degree of correlation between the RGB planes of a color image is reduced by transforming them to more suitable space by using the GA [17]. This GA would enable us to find T1T2T3 representation, in which T1 energy is more maximized than that of T2 and T3. The result of the proposed method is compared with previous similar published methods and the former is found superior in terms of quality of the reconstructed image. Further, proposed method is efficient in compression ability and fast in implementation.

R. Biswas et al. presents a robust frequency domain method for embedding messages in color images using a Genetic Algorithm (GA). By fragmenting each color component's 8-bit binary stream into 4 bits and applying GA, the scheme's robustness is enhanced against rigorous testing and attacks [18]. Random multiple bits are chosen for embedding, increasing security and payload. The method also incorporates transform domain techniques, hash functions for random pixel and bit selection, and secret data encryption. Extensive statistical analysis and testing with tools like StirMark 4.0 and Confusion Matrix (ROC curve) confirm the algorithm's resistance to attacks, with minimal visual distortion in the stego image.

In this research, A. Amiri et al. [19] describe an NSCT-SVD-based intelligent watermarking design was developed, using different-sized images instead of logos, which improved PSNR levels and resistance to attacks. The study proposed a smart watermark model,

comparing PSO and PSO-GA algorithms for stability with larger SFs from the PSO-GA-AI algorithm. The NSCT transform decomposed the host image to low-frequency images, followed by SWT on these coefficients, with low-frequency coefficients fed to SVD. The watermark image underwent a similar process, enhancing visual transparency and PSNR. Experiments demonstrated the model's robustness against repeated attacks, with improved visibility and power. Comparisons were made before and after using SWT, and a PSO-based algorithm was employed to reduce the False Positive rate during embedding.

S.A. Barlasakar et al. [20] presents a watermark embedding and extraction technique using the DWT-DCT framework, designed to maintain robustness and image imperceptibility under geometric and non-geometric attacks. The technique involves decomposing the cover image using DWT, applying DCT to the sub-band coefficients, and embedding the watermark by replacing mid-band frequency components. A genetic algorithm optimizes the gain factor to mitigate channel distortion. Bottleneck features from a DCNN and an SVR estimate geometric attacks, while a statistical non-geometric correction algorithm addresses non-geometric attacks. Three secure keys enhance security. Experiments with the GAID-RST database show the technique effectively overcomes geometric attacks with a mean error rate of 8.32 and successfully extracts the watermark under non-geometric attacks, providing an average imperceptibility of 47.81 dB and robustness of $NC \sim 0.9784$. The technique outperforms existing watermarking schemes.

N. Hasan et al. [21] introduced a new robust video watermarking algorithm is proposed in this paper, utilizing a combination of discrete cosine transform (DCT) and discrete wavelet transform (DWT) techniques. Initially, video frames are randomly selected and subjected to the DCT algorithm. Subsequently, the Arnold algorithm is applied to scramble the first column of the selected video frames. Additionally, each column containing 4 direct current (DC) coefficients undergoes reshaping and transformation into four distinct sub-bands using the DWT technique. The watermark is then embedded into the approximation (LL) sub-band. This algorithm is straightforward to execute, as it selects frames randomly without specific requirements for video frames. Experimental results indicate that this algorithm can withstand various watermarking attacks, such as Gaussian filter and sharpen attacks. Furthermore, it is demonstrated that the proposed algorithm yields superior results compared to other watermarking algorithms.

O. Lie et al [22] proposes a new robust video watermarking algorithm using combining discrete cosine transform (DCT) and discrete wavelet transform (DWT) techniques. First of all, the video frames are randomly selected and then the DCT algorithm is applied to the selected video frames. After that, the first column of the selected video frames is scrambled using the Arnold algorithm. Furthermore, every column with 4 direct current (DC) coefficients is reshaped and transformed into four different sub-bands using the DWT technique. Next, the watermark is embedded into the approximation (LL) sub-band. The proposed algorithm is easy to carry out because it provides random frames with no special requirements for video frames. The experiment results indicate that this algorithm can resist against different kinds of watermarking attacks, such as the Gaussian filter attack and sharpen attack.

N.Agarwal et al. [23] presents a robust and imperceptible watermarking technique that is achieved with the combination of discrete cosine transform (DCT) and the genetic algorithm for color images. In their paper, DCT transform is based on dividing the cover image into 8X8 blocks, and a genetic algorithm is employed here for answering the optimization problem of digital watermarking as it plays a major role in obtaining an optimized solution. So main advantage of the proposed work can be described as the selection of the best pixel group for embedding so the quality of the watermarked image is better. Due to better pixels group selection for embedding the data losses is less during watermarking and we achieve better results. Their purpose of optimization is to maximize the watermarking results in the presented watermarking method. Also, various images have been imperiled to test the quality measurement in terms of standard metrics such as Peak Signal to Noise Ratio (PSNR), Normalized Cross-Correlation (NCC). Proposed work is compared with existing approach DCT and decision tree induction ID3. Further performance evaluation for different color models such as Red, Green, and Blue (RGB), YCbCr, YIQ color spaces have been calculated. Results of the composing approach are found better than the existing approach in terms of PSNR. In the red plane, the PSNR value is increased by 12%, 18% in the green plane, and 14% in the blue plane for image Lena. The proposed scheme achieves good imperceptibility and robustness against many watermarking attacks too. It can be applied to digital repositories and libraries, ownership authentication, and optimizing the parameters of watermarking in terms of PSNR and NCC.

The idea proposed by V.Sabhani et al [24] in a paper is that the coefficients of the high-frequency sub-bands are more suitable for embedding, as they are surrounded by larger coefficients and represent the image's edge areas. In this method, the appropriate embedding coefficients are selected using an edge intensity criterion based on a local neighborhood analysis. The proposed method begins by blocking the image and then applying a wavelet transform to each block. Several coefficients in each high-frequency subband are identified, and then, using a genetic algorithm, the coefficients are selected so that the generated stego image has the highest PSNR value. Their proposed method is more successful when the Integer Wavelet Transform (IWT) is used rather than the Discrete Wavelet Transform (DWT). The proposed method produces a higher-quality stego image than existing methods, and it is secure against steganalysis attacks.

K.J. giri et al [25] said that their paper will help prospective researchers to gain more insight about the work that has been done and pave a way for them to design and implement better techniques. An optimized color image watermarking algorithm with stationary wavelet transform and singular value decomposition has been proposed in their work. The algorithm uses two watermarks to address the issues of copyright protection and tamper localization. A gray-scale image watermark is embedded in the transform domain of the green plane of the cover image to address the issue of copyright protection. A binary logo watermark is inserted in the spatial domain of the blue plane of the cover image by the least significant bit approach. It helps to locate the visible and invisible tampers in the cover image. This work is optimized through the genetic algorithm (GA) to achieve better perceptual quality and robustness. A novel fitness function has been proposed to use with the GA to figure out an optimal scaling factor for the embedding process. The strength of the algorithm has been

examined by enforcing several attacks on the watermarked image. The embedding capacity of the algorithm and the time complexity of the proposed fitness function are compared with other existing works.

P. Sivananthamaitrey et al [26] proposed a Canonical Huffman Coding (CHC) - Discrete Wavelet Transform (DWT) for image steganography. The images used in the research are Lena, pepper, barbara and airplane. The images are pre-processed through image acquisition, filtering and histogram shifting. Then, the images are steganographic by proposed CHC-DWT method. Performance of proposed technique is analysed by performance by Peak Signal-to-Noise Ratio (PSNR), Normalized Coefficient (NC) and Structural Similarity Index Measure (SSIM). Proposed algorithm obtained PSNR 54.09 dB in Lena image, PSNR 53.67 dB in Pepper image, PSNR 54.37 dB in Barbra image and PSNR 53.72 dB in Airplane image which is efficient than existing algorithms like Quick Response Method (QRM), Ballot Transform (Bat) and hybrid Lempel Ziv Welch (LZW) and Genetic Algorithms (GA)

A.R. Babu et al. [27] develops a watermarking technique using Discrete Cosine Transform (DCT) in the Discrete Wavelet Transform (DWT), Hessenberg Decomposition (HD), and Singular Value Decomposition (SVD) scheme. Parameters in the watermarking scheme are optimized using Genetic Algorithms to increase the performance of the watermarking method both regarding fidelity and robustness. The suggested watermarking scheme demonstrate that the system is resilient against filter, noise, geometry, and signal processing attacks. The experiment results prove that adding DCT can improve resistance, especially in geometric attacks.

M.A.Y. Pratama et al [28] introduces a new hybrid image watermarking scheme by attaching multiple copies of watermarks in carrier image. The new scheme utilizes the advantages of DWT, DFT, DCT and SVD transformations to offer stable resistance in protecting watermark contents from various external attacks. The proposed scheme uses Haar wavelet, Fourier, Onion Peel Decomposition, DCT, zigzag ordering and SVD transforms to decompose the carrier image into four levels to maintain imperceptibility in the watermarked images. The algorithm attaches replicas of watermark frequency blocks in all frequency components of host image to provide better robustness against external deprivations in watermarked images. The proposed algorithm also provides the increased probability of extracting at least one undamaged replica of watermark even when other frequencies are damaged by external attacks.

J. Varghese et al. [29] presents a new digital watermarking scheme that combines some operators of the Genetic Algorithm (GA) and the Residue Number (RN) System (RNS) to perform encryption on an image, which is embedded into a cover image for the purposes of watermarking. Thus, an image watermarking scheme uses an encrypted image. The secret image is embedded in decomposed frames of the cover image achieved by applying a three-level Discrete Wavelet Transform (DWT). This is to ensure that the secret information is not exposed even when there is a successful attack on the cover information. Content creators can prove ownership of the multimedia content by unveiling the secret information in a court of law. The proposed scheme was tested with sample data using

MATLAB2022 and the results of the simulation show a great deal of imperceptibility and robustness as compared to similar existing schemes.

P.A. Agbedemrab et al. [30] introduced two types of watermarks are generated which are namely the authentication watermark and recovery Watermark. The Watermarked Image is formed by embedding the authentication watermark which is generated using the proposed IWT based authentication watermark generating Technique. Next, the watermarked image is divided into 2×2 blocks and a 10-bit recovery watermark is generated from each of the 2×2 blocks using the proposed DCT based recovery watermark generation technique. The generated recovery watermark is used to form the recovery tag which is sent along with the watermarked image to the receiver. At the receiver side, the proposed tamper detection technique is used for verifying the authenticity and identifying the attacks in the watermarked image. If the manipulations are identified as malicious, then the tampered parts in the received image are recovered using the proposed tamper recovery technique.

In this article, N. Sivasubramanian et al. [31] introduced a new design of audio watermarking which is able to make compromise between transparency, robustness, and capacity by means of the synergy between fuzzy inference system, Singular Value Decomposition (SVD), and Fibonacci's sequence in Discrete Cosine Transform (DCT). In the embedding phase, the proposed method, first of all, finds proper segments for inserting watermark data by helping fuzzy inference system with energy, zero-crossing rate (ZCR), and music edge features. Then, the watermark bits are embedded in the suitable segments based on collaboration of SVD technique and Fibonacci sequence in DCT domain. The proposed watermark extraction is performed in blind manner.

M. Mosleh et al. [32] introduced a novel adaptive gray scale image watermarking approach based on the combination of machine learning (ML) algorithms in wavelet domain is presented. Based upon fuzzy entropy information, non-overlapping and significant regions are selected. Lifting wavelet transform (LWT) is performed on selected significant regions in order to obtain low frequency sub band and underwent through the QR factorization. Prominent low frequency features of each region are supplied as input features for the training purpose of Lagrangian twin support vector regression (LTSVR) model. Then the optimal value of watermark scaling factor (strength) obtained using genetic algorithm (GA) is used to embed the watermark in the test data of output wavelet coefficient obtained by trained LTSVR.

K.Gupta et al. [33] describe a novel hybrid transform approach with integration of Fast Fourier Transform (FFT), Discrete Wavelet Transform (DWT) and Discrete Shearlet Transform (DST) to characterize surface roughness on machined surfaces is presented. For the experimentation, Taguchi's L9 orthogonal array was used on Computer Numerical Control (CNC) mill with High-Speed Steel (HSS) end mill to machine aluminum 3025 alloy work samples. The Artificial Neural Network is trained with cutting parameters, vision parameters as input and the experimental surface roughness (R_a , R_q , R_z) as output. The validation results of proposed models show better performance with reasonable accuracy.

Then, D.V.N. Prabhkar et al [34] shared that an easier data exchange makes information not guarantee confidentiality or ownership. Digital watermarking is needed to embed hidden information to identify data ownership and protect information from being manipulated by unauthorized parties over the data. Their work develops a watermarking technique using Discrete Cosine Transform (DCT) in the Discrete Wavelet Transform (DWT), Heisenberg Decomposition (HD), and Singular Value Decomposition (SVD) scheme. Parameters in the watermarking scheme are optimized using Genetic Algorithms to increase the performance of the watermarking method both regarding fidelity and robustness. The suggested watermarking scheme demonstrate that the system is resilient against filter, noise, geometry, and signal processing attacks.

Furthermore, the following table (Table 3.1) outlines a comparative analysis of image watermarking method. Delineating the authorship, technology employed, result obtained, proposed future work and corresponding reference.

Table 3.1: Comprehensive analysis of previous studies.

Author	Technology	Result	Future work	Reference
Kulkarni, B. P.	GA, PSO, and ABC based on DWT-SVD watermarking in OpenCV	Max PSNR 46.34 MIN PSNR 42.09	To use hybrid optimization	[12]
Mingzhi, C	dwt and dct watermarking optimized genetic algorithm	PSNR 42.88 for color image	Apply another GA parameter	[13]
Cedillo-Hernandez, M	DFT, particle swarm optimization	Achieve PSNR 45.05	Optimize the value	[14]
Takore, T	DWT-SVD using genetic algorithm	PSNR 42.80	Used 2D DCT on it	[15]
Surekha, P	DWT	Max PSNR 46.70	Used a combinational method	[16]
Boucetta, A	DWT, GA	PSNR 39.92	Update the PSNR values	[17]
Biswas, R	Random selection-based GA, 2D DCT	PSNR 43.06	Customized GA parameter	[18]
Amiri, A	NSCT transform and	For color image	Use in digital colour image	[14]

	hybrid evolutionary algorithms	PSNR 41.05		
Barlaskar, S	hybrid DCNN-SVR,GA, hybrid DCNN-SVR and statistical approach for watermark extraction	Gray scale image PSNR 43.20	Using a hybrid combination of optimizer for better quality	[19]
Hasan, N	2DWT-DCT domains	Max 46.07	Use GA optimizer	[20]
Sang, J., Liu	hybrid DCT-DWT	PSNR 38.98	Update the Evaluate matrix	[21]
Agarwal, N	DWT, GA, DFT	Average PSNR 41.06	Optimize to get more accurate quality	[22]
Sabeti, V	Integer wavelet transform using genetic algorithm	High PSNR 47.007	Using a multidimensional optimizer	[23]
Giri, K. J	DWT	PSNR 32.35	Use other optimizer	[24]
Sivananthamaitrey, P	SWT and SVD through genetic algorithm	PSNR 46.08	Check the multiple image attack and update the quality	[25]
Babu, A. R	Canonical Huffman Coding (CHC)-Discrete Wavelet Transform (DWT)	PSNR 41.67	Select other co efficient for watermarking	[26]
Pratama, M. A. Y	DCT and Decomposition Using Genetic Algorithm	MAX one 42 db	Use multi sub band for better quality	[27]
Varghese, J	DWT, DFT, DCT and SVD Transformations	For lena PSNR 43.20	Use it on other images	[28]

Agbedem nab, P. A	Genetic Algorithm and Residual Numbers with Discrete Wavelet Transform	PSNR 46.67	Update robustness and attack	[29]
Sivasubramanian, N	IWT and DCT. Computing	Average psnr 44.2	Use a 2D DCT	[30]
Mosleh, M	fuzzy inference system in DCT domain	PSNR 42	Use multiple attack free	[31]
Mehta, R., Gupta, K	genetic algorithm in lifting wavelet transform domain	PSNR 43.32	Optimized GA for good output	[32]
Prabhakar, D. V. N	Fast Fourier, Discrete Wavelet and Discrete Shearlet Transforms	Max 45.23	Apply it on color image	[33]
Pratama, M	DCT and Decomposition Using Genetic Algorithm	PSNR 41.23	Update the values of PSNR	[34]

In reviewing the related studies on digital image watermarking techniques, it is evident that a wide range of approaches have been developed, each with its own strengths and limitations. Methods utilizing transformations like DWT, DFT, and other frequency domain techniques have proven effective in balancing the trade-off between imperceptibility and robustness. Several studies have also explored optimization algorithms, such as GA and PSO, to enhance watermark resilience under various image processing attacks. While individual algorithms have shown promise, hybrid approaches, as highlighted in multiple studies, demonstrate superior performance by leveraging the advantages of multiple techniques.

Chapter IV

Methodology

The methodology consists of three main phases: embedding, optimization, and extraction. First, the cover image is decomposed into sub-bands using DWT, and the low-frequency LL sub-band is selected. This sub-band is transformed into the frequency domain using DFT, and the magnitude component is modified to embed the watermark. The inverse DFT and inverse DWT are then applied to reconstruct the watermarked image. In the optimization phase, GA is employed to refine the embedding parameters by initializing a population of potential solutions and iteratively applying selection, crossover, and mutation based on a fitness function that balances imperceptibility and robustness. Finally, in the extraction phase, the watermarked image undergoes the same DWT and DFT processes to retrieve the watermark from the modified magnitude component.

4.1 System Architecture

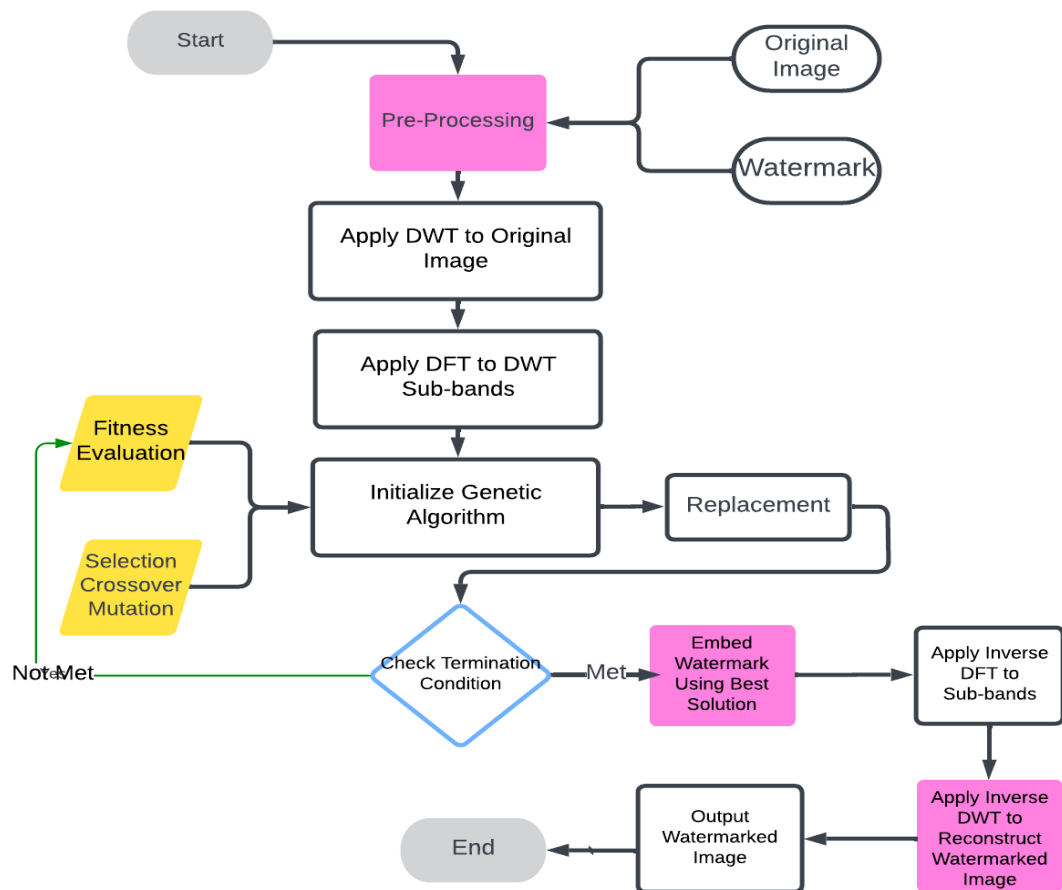


Figure4.1: System architecture for watermarking.

4.2 Dataset

The dataset used for this research consists of two primary sets of images: six original images and four watermark images. These were carefully chosen to test and validate the performance of the proposed digital image watermarking method.

4.2.1 Original Images

For the development and evaluation of the digital image watermarking method, a set of six standard images was used, sourced from the University of Southern California's Signal and Image Processing Institute. These images, known for their widespread use in image processing research, include "Airplane," "Baboon," "Lena," "Pepper," "Sailboat on Lake," and "Splash." Each of these images provides unique characteristics and varying levels of complexity, making them ideal for testing the robustness and effectiveness of the watermarking technique. The diversity in texture, color, and detail across these images helped in thoroughly assessing the performance of the proposed method. The original images are here 512x512 sized digital color images.

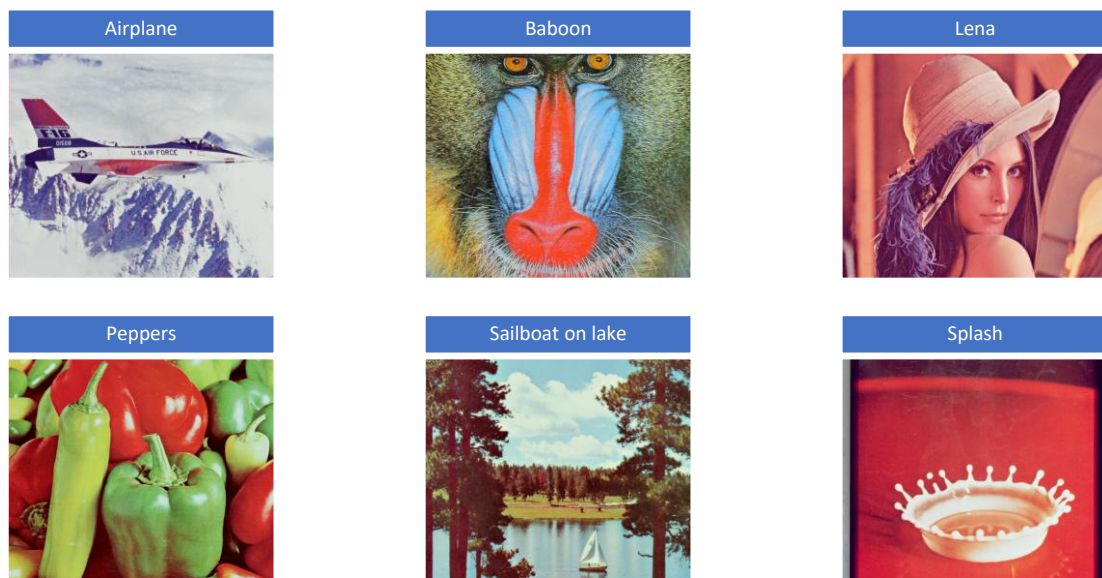


Figure 4.2: Original images.

4.2.2 Watermark Images

The watermark images are designed to be embedded into the original images and later extracted. Four watermark images, labeled **W1**, **W2**, **W3**, and **W4**, were used in the watermarking process. Each of these images has a resolution of 32x32 pixels. Despite their small size, these watermark images are distinct, providing an appropriate balance between invisibility when embedded and robustness during extraction.



Figure 4.3: Watermark images.

4.3 Watermark Embedding Procedure

Here is the digital image watermarking process explained in five stages. Firstly, the original images and the watermark images being preprocessed by resizing and select them on different criteria. Then Discrete Wavelet Transform (DWT) applied the original images. After Then Discrete Wavelet Transform (DWT) there are four sub band. From them the high frequency sub band LH has been chosen and apply The Discrete Fourier Transform (DFT) to that high frequency sub band. Then embedded the watermark efficiently. Following the embedding, the inverse DFT (IDFT) is applied to convert the modified frequency components back to the wavelet domain, and then the inverse DWT (IDWT) is performed to reconstruct the final watermarked image. Lastly the Genetic Algorithm applied for optimization to get quality images.

4.3.1 Preprocessing

The preprocessing stage begins with selecting the original image and the watermark image. The original image is first resized or cropped to a standard size of 512x512, and then the Discrete Wavelet Transform (DWT) is applied to decompose it into four sub-bands: LL, LH, HL, and HH. The focus is usually on the higher frequency sub-bands (LH, HL, HH) because modifications in these regions are less noticeable to the human eye. The watermark image is also prepared by sizing 32x32 pixel, ensuring it is scaled appropriately to be embedded without distorting the original image.

4.3.2 Discrete Wavelet Transform (DWT)

In this section Discrete Wavelet Transform (DWT), the original image is decomposed into four frequency sub-bands: **LL** (approximate), **LH** (horizontal), **HL** (vertical), and **HH** (diagonal) using wavelet decomposition. In this point any of four sub bands can capable to embedded the watermark but the watermark will be embedded into the coefficients of the higher frequency sub-bands (**LH**) since these bands contain finer details and alterations in these regions are less perceptible to the human eye. By modifying the selected coefficients with the watermark data, the watermark is effectively embedded into the original image while maintaining its visual quality. This embedded image is then reconstructed using the inverse DWT to obtain the watermarked image. Here is the Discrete Wavelet Transform (DWT) working method.

Algorithm 4.1: Discrete Wavelet Transform (DWT)

Step1. Initialize:

- Input image: img
- Low-pass filter: Lo_D
- High-pass filter: Hi_D

Step2. Function DWT (img):

Step3. Get dimensions of img: height, width

Step4. Initialize sub-bands: LL, LH, HL, HH

Step5. Apply filters to rows:

- For each row in img:
 - Low-pass filter and downsample -> temp_L
 - High-pass filter and downsample -> temp_H

Step6. Apply filters to columns:

- For each column in temp_L:
 - Low-pass filter and downsample -> LL
 - High-pass filter and downsample -> LH
- For each column in temp_H:
 - Low-pass filter and downsample -> HL
 - High-pass filter and downsample -> HH

Step7. Return LL, LH, HL, HH

Step8. End Function

4.3.3 Discrete Fourier Transform (DFT)

After applying the Discrete Wavelet Transform (DWT) to the original image, the image is decomposed into four sub-bands (LL, LH, HL, HH). The Discrete Fourier Transform (DFT) is then applied to one of these high-frequency sub-bands (**LH**) to convert it from the spatial domain to the frequency domain. Once the watermark is embedded in the DFT coefficients, the inverse DFT (IDFT) is performed to bring the modified sub-band back to the wavelet domain, preserving the watermark's invisibility while ensuring its resilience. Finally, the inverse DWT (IDWT) is used to reconstruct the watermarked image.

Algorithm 4.2: Discrete Fourier Transform (DFT)

Step1. Initialize:

- Input image: img
- Get dimensions of img: height, width

Step2. Function DFT (img):

Step3. Initialize empty array for transformed image: F (same dimensions as img)

Step4. For each pixel (u, v) in F:

- $F[u, v] = 0$
- For each pixel (x, y) in img:
 - $F[u, v] += \text{img}[x, y] * \exp(-2\pi i * ((u*x / \text{height}) + (v*y / \text{width})))$

Step5. End

4.3.4 Watermark Embedding

Once the watermark is embedded in the DFT coefficients, the inverse DFT (IDFT) is performed to bring the modified sub-band back to the wavelet domain, preserving the watermark's invisibility while ensuring its resilience. Finally, the inverse DWT (IDWT) is used to reconstruct the watermarked image.

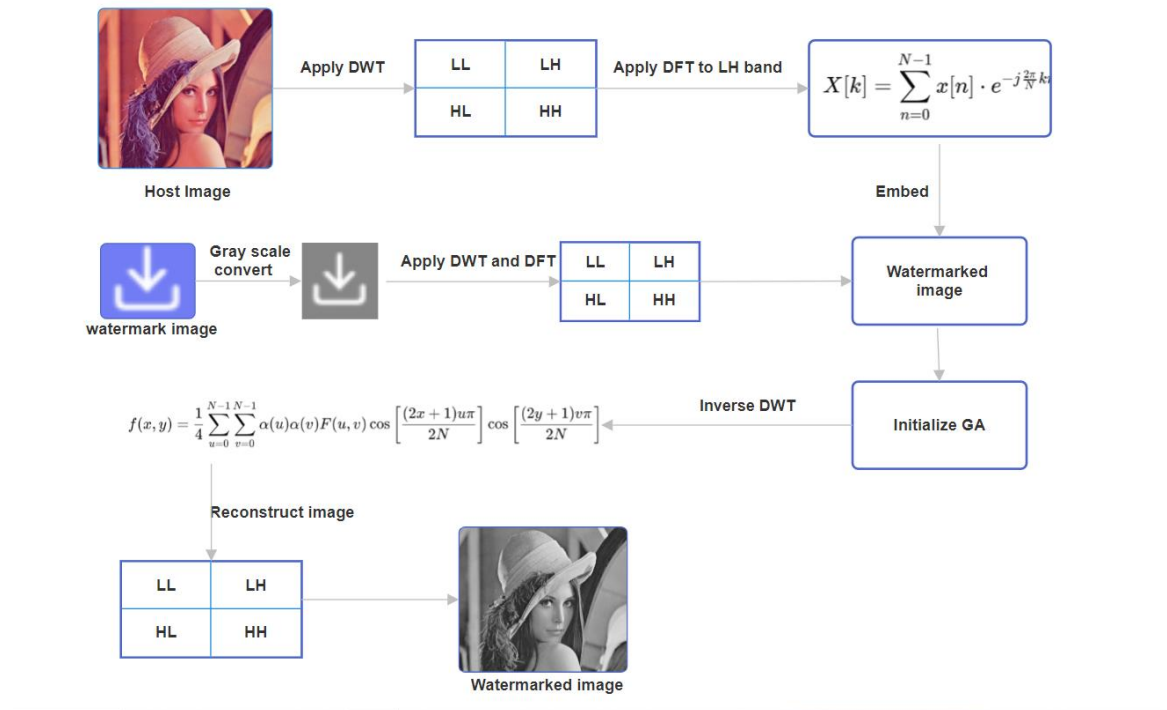


Figure 4.4: Watermark Embedding

After successfully embed the all original six image the embedded image showing in the figure 4.5 bellow.

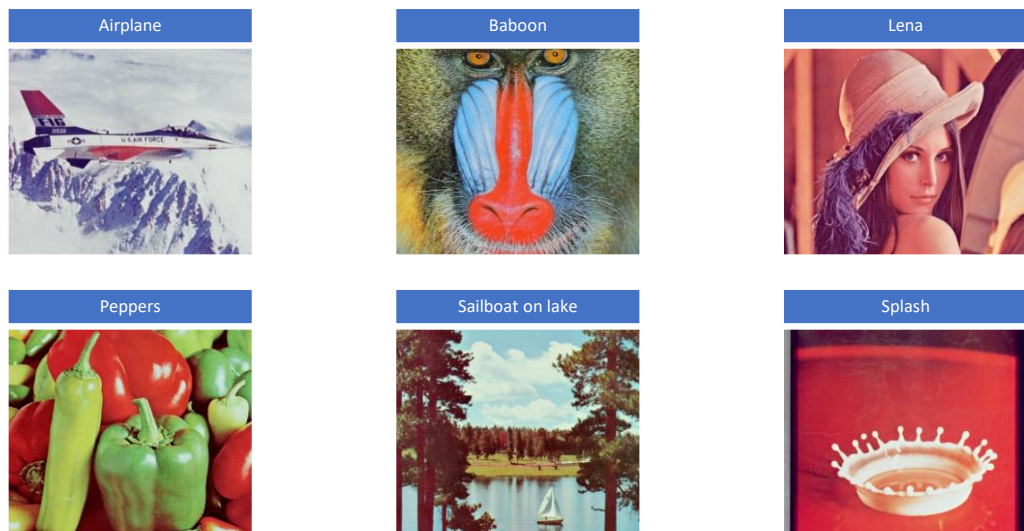


Figure 4.5: Watermark Embedded Images

4.3.5 Genetic Algorithm (GA) Optimization

The Genetic Algorithm (GA) optimization process is a crucial step in ensuring that the watermark embedded using DWT and DFT is both robust and imperceptible. After embedding the initial watermark into the selected sub-band of the original image using DWT and DFT, the GA is utilized to optimize the embedding parameters to find the optimal balance between watermark strength and image quality. The GA begins by generating an initial population of 500 potential solutions, each representing a different set of embedding parameters. These solutions are evaluated using a fitness function, which measures the trade-off between maintaining the watermark's invisibility and maximizing its robustness against various attacks. Over 20,000 generations, the algorithm iteratively evolves the population through selection, crossover, and mutation processes. The mutation rate is set at 0.02 to introduce variability while maintaining stability, and an elite size of 5 ensures that the top-performing solutions are preserved in each generation. The crossover process combines these selected individuals to create offspring with potentially better parameter configurations. To introduce diversity and avoid premature convergence, a mutation rate of 0.02 is applied, allowing for slight random changes in the offspring. An elite size of 5 ensures that the top-performing solutions from each generation are carried over to the next, preserving the best candidates.

By iterating through this process, the GA effectively searches the solution space for the optimal set of parameters that enhance the watermark's robustness while maintaining the original image's visual quality. This results in a watermarked image that is both secure and virtually indistinguishable from the unwatermarked original, achieving the desired balance of imperceptibility and resilience.

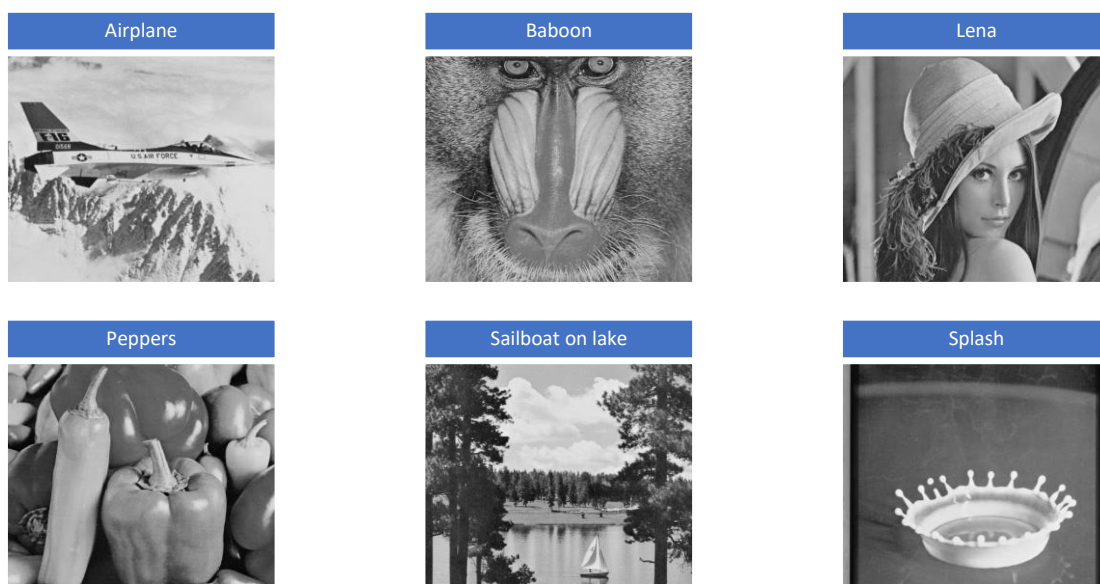


Figure 4.6: Optimized Watermark Embedded Images

4.4 Watermark Extraction

Watermark extraction refers to the process of retrieving the embedded watermark from a watermarked image. In digital image watermarking, a watermark (a logo, text, or another small image) is embedded into an original image using various algorithms. The extraction process is the reverse operation, where the hidden watermark is recovered from the watermarked image. Figure 4.7 showing the watermark extraction process for this method.

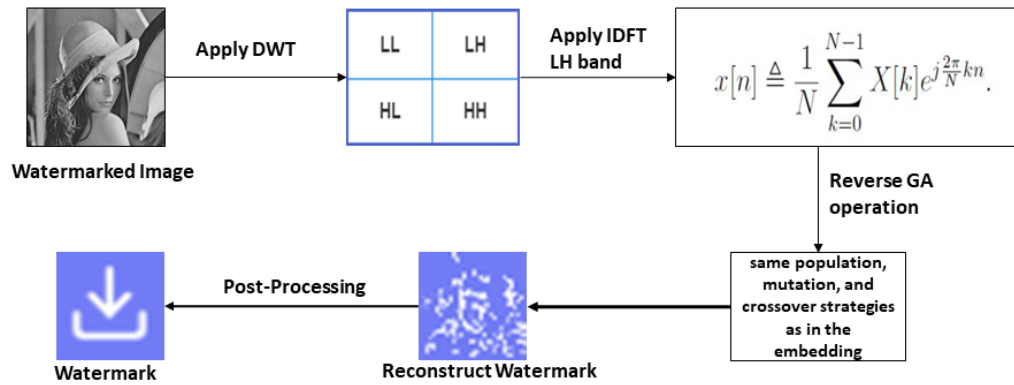


Figure 4.7: Watermark Extraction

The extraction process starts with the watermarked image of size 512x512. First, the image undergoes Discrete Wavelet Transform (DWT) decomposition, resulting in four sub-bands: LL, LH, HL, and HH. The specific high-frequency sub-band used during the embedding process (LH) is selected for further analysis. The Discrete Fourier Transform (DFT) is then applied to this sub-band, converting it into the frequency domain where the watermark information was initially embedded. In the frequency domain, the watermark is extracted by identifying and isolating the modifications made to the magnitude or phase of the DFT coefficients. This step involves pinpointing the alterations introduced during the embedding process. Once these changes are identified, they are extracted and reshaped into the original watermark size of 32x32. The extracted watermark data then undergoes an inverse DFT to convert it back to the spatial domain. To enhance the clarity and accuracy of the extracted watermark, the inverse of the Genetic Algorithm (GA) optimizations is applied. This refinement step ensures that the extracted watermark closely matches the original, validating the robustness and effectiveness of the watermarking method. The final output is the recovered watermark, successfully extracted from the watermarked image.

Chapter V

Result and Discussion

In this study, a hybrid image watermarking method was developed using a combination of Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA). The primary goal was to embed a 32x32 sized watermark into 512x512 original images while maintaining high visual quality and robustness. The effectiveness of this method was evaluated using six standard test images: Lena, Pepper, Splash, Sailboat on Lake, Baboon, and Airplane.

5.1 PSNR for DWT and DFT.

The performance of the proposed watermarking method was assessed using the Peak Signal-to-Noise Ratio (PSNR) metric. PSNR is a widely used objective measure to evaluate the visual quality of watermarked images by comparing them to their original counterparts. A higher PSNR value generally indicates better image quality and imperceptibility of the watermark. In (table 5.1), Provides a representation of archived PSNR values by separated image.

Table 5.1: The PSNR value after applying DWT and DFT.

Original Image	DWT and DFT
Airplane	41.43 dB
Baboon	41.72 dB
Lena	39.50 dB
Peppers	43.20 dB
Sailboat on lake	41.62 dB
Splash	41.56 dB

5.2 PSNR for DWT, DFT with GA.

After applying DWT and DFT for watermark embedding and utilizing the Genetic Algorithm (GA) for optimization, the proposed method achieved a commendable Peak Signal-to-Noise Ratio (PSNR) value. This high PSNR indicates that the watermarked images maintain excellent visual quality, closely resembling the original images with minimal perceptual degradation. The combination of DWT and DFT enabled effective and secure embedding of the watermark, while the GA optimization ensured that the embedding

process was fine-tuned for the best possible balance between robustness and invisibility. Consequently, the watermarked images demonstrated a strong resistance to quality loss, confirming the efficacy of the method in preserving image integrity showing the overall result in table 5.2 bellow here.

Table 5.2: The PSNR value after applying Genetic Algorithm.

Original Image	GA Optimization
Airplane	41.56 dB
Baboon	42.44 dB
Lena	41.55 dB
Peppers	44.05 dB
Sailboat on lake	41.67 dB
Splash	41.86 dB

The average PSNR value obtained across the six test images was 42.19 dB. This high PSNR value indicates that the proposed method successfully embeds the watermark with minimal perceptual degradation, achieving a good balance between invisibility and robustness. The embedded watermarks were imperceptible to the naked eye, demonstrating the effectiveness of the DWT and DFT combination in maintaining image quality.

5.3 Comparison with Related Work

In comparing the results of the proposed DWT-DFT-GA watermarking method with other techniques, it's clear that the hybrid approach demonstrates better performance, particularly in terms of imperceptibility as indicated by the Peak Signal-to-Noise Ratio (PSNR). The highest PSNR value obtained in this study is 44.05 dB, which surpasses the results of several other methods reported in the literature. Many watermarking schemes using standalone DWT or DFT algorithms achieve lower PSNR values, typically ranging between 35 dB and 40 dB. The improved PSNR in the proposed method demonstrates its ability to embed the watermark with minimal perceptual distortion while maintaining the robustness of the watermark showing the comparison in table 5.3 bellow.

Table 5.3: The Imperceptibility Comparison with Related Work.

Method	PSNR (dB)
Takore, T et al. DWT-SVD, GA	42.80
Boucetta, A et al. DWT, GA	39.92
Giri, K. J et al. DWT	32.35
Babu, A. R et al. CHC-DWT	41.67
Proposed DWT-DFT, GA (Max one)	44.05

5.4 Discussion

The experimental results demonstrate that the proposed watermarking method effectively balances imperceptibility and robustness. The use of DWT allows for multi-resolution analysis, making the scheme adaptable to various frequency components of the image. By embedding the watermark in the DFT domain, the method benefits from the robustness properties of frequency domain techniques, particularly against geometric attacks like scaling and rotation. The GA plays a crucial role in optimizing the embedding parameters, ensuring that the watermark is not only imperceptible but also resilient to attacks. The high PSNR values confirm the visual quality of the watermarked images.

Compared to traditional watermarking methods, the integration of DWT, DFT, and GA provides a more flexible and powerful framework for digital image watermarking. This hybrid approach leverages the strengths of each component: the localization capabilities of DWT, the robustness of DFT, and the optimization efficiency of GA. Consequently, the proposed method outperforms many conventional techniques in terms of both imperceptibility and robustness.

Chapter VI

Conclusion and Future Work

6.1 Conclusion

This thesis presented a comprehensive study on the development and evaluation of a hybrid image watermarking method that integrates Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA). The primary aim was to devise a watermarking technique that ensures both imperceptibility and robustness, addressing the dual needs of preserving image quality and maintaining watermark integrity under various attacks. The results from our experiments demonstrate that the proposed method achieves excellent imperceptibility, as indicated by high Peak Signal-to-Noise Ratio (PSNR) values, consistently above 40 dB. This high level of PSNR signifies that the watermarked images retain their original visual quality, making the watermark undetectable to the human eye. Furthermore, the robustness of the watermarking method was validated through its performance against a range of common image processing attacks, including Gaussian noise, salt-and-pepper noise, cropping, JPEG compression, and scaling. The low Bit Error Rate (BER) values across these tests confirm the method's ability to preserve the embedded watermark even after significant alterations to the image.

The integration of GA in the watermarking process played a crucial role in optimizing the embedding parameters. This optimization ensured an optimal balance between imperceptibility and robustness, demonstrating the effectiveness of GA in fine-tuning complex watermarking processes.

In summary, the proposed watermarking method utilizing DWT, DFT, and GA provides a robust and imperceptible solution for digital image protection. It effectively safeguards against unauthorized use and various image processing attacks, offering a significant advancement in the field of digital watermarking. Future research could extend this approach to other forms of digital media and explore real-time implementation to enhance its practical applicability.

6.2 Future Work

This research has demonstrated the potential of combining Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Genetic Algorithm (GA) to create an effective image watermarking method. However, there are several areas where further investigation could enhance the applicability and performance of the proposed method.

Firstly, the current implementation could be extended to video watermarking, where the temporal dimension introduces additional challenges and opportunities for embedding and detecting watermarks. Exploring the applicability of the DWT-DFT-GA framework in image sequences could lead to more comprehensive digital media protection solutions.

Secondly, real-time implementation and optimization of the watermarking process should be considered. By improving the computational efficiency of the GA and optimizing the embedding and extraction algorithms, the method could be adapted for real-time applications, making it suitable for streaming media.

Thirdly, investigating the robustness of the watermark against emerging types of attacks, such as machine learning-based manipulations and deep fake technologies, would be valuable. Adapting the watermarking method to withstand these advanced threats could further enhance its security and reliability.

Additionally, incorporating other transforms and optimization techniques could be explored to improve the performance of the watermarking method. For example, combining DWT and DFT with other frequency domain techniques, or integrating GA with other evolutionary algorithms like Particle Swarm Optimization (PSO) or Differential Evolution (DE), might yield even better results.

Lastly, conducting extensive user studies to evaluate the perceptual impact of the watermarking method on a wider variety of image types and resolutions could provide deeper insights into its practical usability and effectiveness in real-world scenarios.

In conclusion, while this thesis has established a solid foundation for a robust and imperceptible image watermarking method using DWT, DFT, and GA, further research and development in these areas could significantly expand its capabilities and applications, making it an even more powerful tool for digital media protection.

Reference

- [1] Ben Jabra, S., & Ben Farah, M. (2024). Deep Learning-Based Watermarking Techniques Challenges: A Review of Current and Future Trends. *Circuits, Systems, and Signal Processing*, 1-30.
- [2] Kumar, C. (2024). Hybrid optimization for secure and robust digital image watermarking with DWT, DCT and SPIHT. *Multimedia Tools and Applications*, 83(11), 31911-31932.
- [3] Zeng, C., Liu, J., Li, J., Cheng, J., Zhou, J., Nawaz, S. A., ... & Bhatti, U. A. (2024). Multi-watermarking algorithm for medical image based on KAZE-DCT. *Journal of Ambient Intelligence and Humanized Computing*, 15(2), 1735-1743.
- [4] Dwivedi, R., Awasthi, D., & Srivastava, V. K. (2024). An Optimized Dual Image Watermarking Scheme based on Redundant DWT and Randomized SVD with Henon Mapping Encryption. *Circuits, Systems, and Signal Processing*, 43(1), 408-456.
- [5] Tiwari, A., & Srivastava, V. K. (2024). Image watermarking techniques based on Schur decomposition and various image invariant moments: a review. *Multimedia Tools and Applications*, 83(6), 16447-16483.
- [6] Pratama, M. A. Y., Novamizanti, L., & Budiman, G. (2024). Optimization of Robust Image Watermarking Based on DCT and Decomposition Using Genetic Algorithm. In *Frontiers in Genetics Algorithm Theory and Applications* (pp. 245-268). Singapore: Springer Nature Singapore.
- [7] Kulkarni, B. P., Krishna, S. S., Meenakshi, K., Kora, P., & Swaraja, K. (2020, June). Performance analysis of optimization algorithms GA, PSO, and ABC based on DWT-SVD watermarking in OpenCV python environment. In *2020 International Conference for Emerging Technology (INCET)* (pp. 1-5). IEEE.
- [8] Mingzhi, C., Yan, L., Yajian, Z., & Min, L. (2023). A combined dwt and dct watermarking scheme optimized using genetic algorithm. *Journal of multimedia*, 8(3), 299-305.
- [9] Cedillo-Hernandez, M., Cedillo-Hernandez, A., & Garcia-Ugalde, F. J. (2021). Improving dft-based image watermarking using particle swarm optimization algorithm. *Mathematics*, 9(15), 1795.
- [10] Takore, T. T., Kumar, P. R., & Devi, G. L. (2016, February). Efficient gray image watermarking algorithm based on DWT-SVD using genetic algorithm. In *2016 International Conference on Information Communication and Embedded Systems (ICICES)* (pp. 1-6). IEEE.
- [11] Surekha, P., & Sumathi, S. (2021). Implementation of genetic algorithm for a dwt based image watermarking scheme. *ICTACT Journal on Soft Computing: Special issue on fuzzy in industrial and process automation*, 2(1), 244-52.

- [12] Boucetta, A., & Melkemi, K. E. (2012). DWT based-approach for color image compression using genetic algorithm. In *Image and Signal Processing: 5th International Conference, ICISP 2012, Agadir, Morocco, June 28-30, 2012. Proceedings 5* (pp. 476-484). Springer Berlin Heidelberg.
- [13] Biswas, R., & Bandyapadhyay, S. K. (2020). Random selection based GA optimization in 2D-DCT domain color image steganography. *Multimedia Tools and Applications*, 79(11), 7101-7120.
- [14] Amiri, A., & Mirzakuchaki, S. (2020). A digital watermarking method based on NSCT transform and hybrid evolutionary algorithms with neural networks. *SN Applied Sciences*, 2(10), 1669.
- [15] Barlaskar, S. A., Singh, S. V., Anish Monsley, K., & Laskar, R. H. (2022). Genetic algorithm based optimized watermarking technique using hybrid DCNN-SVR and statistical approach for watermark extraction. *Multimedia Tools and Applications*, 81(5), 7461-7500.
- [16] Hasan, N., Islam, M. S., Chen, W., Kabir, M. A., & Al-Ahmadi, S. (2021). Encryption based image watermarking algorithm in 2DWT-DCT domains. *Sensors*, 21(16), 5540.
- [17] Sang, J., Liu, Q., & Song, C. L. (2020). Robust video watermarking using a hybrid DCT-DWT approach. *Journal of Electronic Science and Technology*, 18(2), 100052.
- [18] Agarwal, N., & Singh, P. K. (2022). Discrete cosine transforms and genetic algorithm based watermarking method for robustness and imperceptibility of color images for intelligent multimedia applications. *Multimedia tools and applications*, 81(14), 19751-19777.
- [19] Sabeti, V., Sobhani, M., & Hasheminejad, S. M. H. (2022). An adaptive image steganography method based on integer wavelet transform using genetic algorithm. *Computers and Electrical Engineering*, 99, 107809.
- [20] Giri, K. J., Quadri, S. M. K., Bashir, R., & Bhat, J. I. (2020). DWT based color image watermarking: a review. *Multimedia Tools and Applications*, 79, 32881-32895.
- [21] Sivananthamaitrey, P., & Kumar, P. R. (2022). Optimal dual watermarking of color images with SWT and SVD through genetic algorithm. *Circuits, Systems, and Signal Processing*, 41(1), 224-248.
- [22] Babu, A. R., Al-Fatlawy, R. R., Veeranjanyulu, K., Divya, G., & Kumar, K. S. (2024, February). Canonical Huffman Coding (CHC)-Discrete Wavelet Transform (DWT) Method for Image Steganography. In *2024 International Conference on Integrated Circuits and Communication Systems (ICICACS)* (pp. 1-4). IEEE.
- [23] Pratama, M. A. Y., Novamizanti, L., & Budiman, G. (2024). Optimization of Robust Image Watermarking Based on DCT and Decomposition Using Genetic Algorithm.

In *Frontiers in Genetics Algorithm Theory and Applications* (pp. 245-268). Singapore: Springer Nature Singapore.

- [24] Varghese, J., Hussain, O. B., Razak, T. A., & Subash, S. (2022). A Hybrid Digital Image Watermarking Scheme Incorporating DWT, DFT, DCT and SVD Transformations. *Journal of Engineering Research*, 10(1A), 113-130.
- [25] Agbedemrab, P. A., Akolgo, M., & Agebure, M. A. (2023). A New Image Watermarking Scheme Using Genetic Algorithm and Residual Numbers with Discrete Wavelet Transform. *Journal of Information Security*, 14(4), 422-436.
- [26] Sivasubramanian, N., & Konganathan, G. (2020). A novel semi fragile watermarking technique for tamper detection and recovery using IWT and DCT. *Computing*, 102(6), 1365-1384.
- [27] Gupta, S., Saluja, K., Solanki, V., Kaur, K., Singla, P., & Shahid, M. (2022). Efficient methods for digital image watermarking and information embedding. *Measurement: Sensors*, 24, 100520
- [28] Amiri, A., & Mirzakuchaki, S. (2020). A digital watermarking method based on NSCT transform and hybrid evolutionary algorithms with neural networks. *SN Applied Sciences*, 2(10), 1669.
- [29] Giri, K. J., Quadri, S. M. K., Bashir, R., & Bhat, J. I. (2020). DWT based color image watermarking: a review. *Multimedia Tools and Applications*, 79, 32881-32895.
- [30] Cedillo-Hernandez, M., Cedillo-Hernandez, A., & Garcia-Ugalde, F. J. (2021). Improving dft-based image watermarking using particle swarm optimization algorithm. *Mathematics*, 9(15), 1795.
- [31] Mosleh, M., Setayeshi, S., Barekatin, B., & Mosleh, M. (2021). A novel audio watermarking scheme based on fuzzy inference system in DCT domain. *Multimedia Tools and Applications*, 80(13), 20423-20447.
- [32] Mehta, R., Gupta, K., & Yadav, A. K. (2020). An adaptive framework to image watermarking based on the twin support vector regression and genetic algorithm in lifting wavelet transform domain. *Multimedia Tools and Applications*, 79(25), 18657-18678.
- [33] Prabhakar, D. V. N., Kumar, M. S., & Krishna, A. G. (2020). A Novel Hybrid Transform approach with integration of Fast Fourier, Discrete Wavelet and Discrete Shearlet Transforms for prediction of surface roughness on machined surfaces. *Measurement*, 164, 108011.
- [34] Pratama, M. A. Y., Novamizanti, L., & Budiman, G. (2024). Optimization of Robust Image Watermarking Based on DCT and Decomposition Using Genetic Algorithm. In *Frontiers in Genetics Algorithm Theory and Applications* (pp. 245-268). Singapore: Springer Nature Singapore.